1. STUDY OF VARIOUS COMMANDS USED IN LINUX

AIM: To study about various networking commands on Linux OS.

→ pwd : prints current working directory.

output:

```
$pwd
/home/linuxuser
```

→ mkdir : creates a new directory

o/p:

```
$mkdir fedora
$ ls
Desktop documents downloads
fedora
```

→ cd : change current directory.

o/p:

```
$cd fedora
$ pwd
/home/linuxuser/fedora
```

→ touch filename.txt : creates empty file

o/p:

```
$ touch example.txt
$ ls
example.txt
```

→ echo "Hi" > filename.txt

description : write text into a file

o/p:

```
$ echo "Hi" > example.txt
```

→ cat filename.txt : Displays contents of file

o/p:

$ cat example.txt

Hi

→ grep

Desciption: searches for specific pattern in file

o/p:

$ grep "Hi" example.txt

Hi

wc

→ wc

Desciption: Displays the no. of lines, words & bytes in file

o/p:

$ wc example.txt

1 1 3 example.txt

→ wc -l

Desciption: count the number of lines in a file

o/p:

$ wc -l example.txt

1 example.txt

→ wc -w

discription: counts the no. of words in a file.

o/p:

$ wc -w example.txt

→ wc -c

discription: count no. of bytes in a file

o/p:

$ wc -c example.txt

3 example.txt

wc -m.

→ wc-m

count no. of characters in file

O/P:

$ wc-m example

2 example.txt.

→ wc-L

Display length of longest line in file

O/P:

$ wc-L example.txt

2 example.txt.

Basic Networking commands

→ hostname

displays system hostname

O/P:

$ hostname

local host. linu

→ route : display routing table.

O/P: $ route

Kernel IP routing table.

→ nestat : shows network connections & routing information.

O/P: $ netstat

Active internet connection.

→ IP address show : displays IP address & network interface details.

O/P:

$ ip address show

1. lo

2. exp083

→ nslookup : find IP address of domain

O/P: $nslookup www.google.com

name : www.google.com.

address : 142.250.xxx.xxx

→ ipconfig

O/P:

$ipconfig

exp083

inet 10.22.15

→ ping

O/P:

$ Ping www.google.com

64 bytes from.

→ arp : display ARP cache

O/P:

$arp -a

-gateway (10.0.2.2)

→ traaroute

O/P:

$traceroute www.google.com

1. gateway.

2. ....

Result

Thus various networking commands on Linux operating system is successfully verified & executed.

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2. Manual Assignment of IP Address for WIFI connection (Windows)

AIM: To configure & assign an IP address manually to a WIFI adaptor in windows OS.

Procedure:

1. open control pannel.
2. click change adapter settings on left side
3. Right click on wifi adaptor properties.
4. Select internet protocol version 4 (TCP / IPV4) & click property
5. select following IP address & enter
   - * IP address 192.168.1.20
   - * Subnet : 255.255.255.0
   - * Default gateway : 192.168.1.1
   - * prefered DNS server : 8.8.8.8
6. click OK & close all windows.
7. open command prompt & verify configuration

   Name: ipconfig

   ping 192.168.1.1

   ping www.google.com.

command prompt o/p:

C:\> ipconfig

windows ip configuration

windows LAN adaptor WIFI

IPV4 address. .

192.168.1.20.

subnet Mask.

255.255.255.0

Default gateway . . . 192.168.1.1

c:\> ping 192.168.1.1

Reply from 192.168.1.1

Reply from 192.168.16

time <1 ms TTL = 64

Reply from 192.168.1.1 bytes = 32

time <1 ms TTL = 64.

c:\> ping www.google.com

Reply from www.google.com: bytes = 32

time = 25 ms TTL = 117

Reply from www.google.com: bytes = 32

time = 24 ms TTL = 117

Result :

The WIFI adapter was successfully configured with a manual IP address. The computer was able to communicate with routes & access internet successfully.

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3. Study of Different types of Network cables & crimping of cable with RJ45 connector.

AIM: To study different types of network cables & to prepare a network cable using RJ45 connectors for establishing LAN connections b/w network devices.

Apparatus required:

1.	CTD 5e / UTP cable	1 roll
2.	RJ45 / (8P8C)	2 nos
3.	crimping tool	1
4.	cable / other	1
5.	computer system	as required.

Theory:

A network cable is medium through with data & instructions b/w computer & other devices.

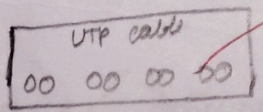
1. Types of network cables

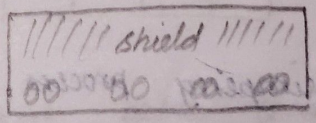
① Twisted pair cable:

* most commonly used for LAN
* consist of pair of insulated copper wire twisted.

types:

* UTP
* no metallic shielding used in LAN.





category	speed	frequency
CAT 5	upto 100 mbps	100 MHz
CAT 5e	upto 1 gbps	100 MHz
CAT 7	upto 10 Gbps	600 MHz

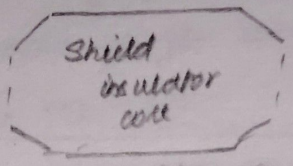
2) coaxial cable

* has central copper conductor, dielectric insulator
* used in cable TV

types:

Thin net (10 base 2)

Thick net (10 base 5)



3) Fibre optic cable:

* uses light signals instead of electrical signals.

types:

* single mode fibre (SMF)
* multi mode fibre (MMF)

4) RJ45 connector

* RJ - Registered Jack.
* RJ45 is 8 pin modular connector.

each pin carries specific signal in network.

1	2	3	4	5	6	7	8

5) crimping process

a) For straight through cable.

b) For crossover cable.

LAN	JF	STAR
1	2	3	4
5	6	7	8

Procedure:

1) cut the UTP cable

2) strip cable.

3) untwist & arrange pairs

4) Trim the ends.

5) Insert wires into RJ45 connector

6) crimp the connector

7) Repeat other end.

8) Test cable.

Observation table

cable type	end 1 code	end 2 code	Purpose	Result
Straight	T568B	T568B	PC-switch	working
cross over	T568B	T568B	PC-PC	working

Result: Different types of network cables connect & studied successfully.

A UTP CAT5 cable was crimped using RJ45 & straight to straight & cable connectivity b/w network was successfully established & Tested.

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4. PACKET TRACER TOOL INSTALLATION AND USER INTERFACE

AIM:

TO study the cisco Packet Tracer tool, its installation, user interface & analyse the behaviour of network devices using the Packet Tracer simulator

Introduction:

Packet Tracer is a network simulation & modeling tool developed by cisco. It allows users to design, configure troubleshoot computer networks without requiring physical network devices. It is available for windows & linux platforms & helps users understand networking concepts through simulation. The protocols in Packet Tracer behave similarly to real networking hardware.

Installing Packet Tracer:

Download Packet Tracer from CISCO Networking Academy & install it using setup file for windows or the installer for Linux

Follow the installation steps to complete the setup.

User Interface Overview:

The packet tracer interface included the menu bar, main tool bar, Logical / Physical workspace.

common tools Bar, Real time / simulation tabs, Network components Box and user created packet Box.

These components help create, configure & simulate network topologies.

Analyse the behaviour of Network devices using CISCO Packet tracer.

click on copper straight-through cable, select the PC & connect it to hub using the cable. The Link is up.

Similarly connect remaining 3PCs to the HUB. Similarly connect 4PCs to the Switch using copper straight through cable.

RESULT:

Thus, Packet Tracer tool, its installation, user interface & to understand the CISCO Packet Tracer environment for designing simple networks is studied & implemented.

5 Develop a customised ping command to test the server connectivity.

Aim: To develop a customised ping command to test the server connectivity.

Ping command is networking utility used to test whether a device / website is reachable over a network. It also measures response time.

commands:	Description
ping google.com	ping a website
ping 8.8.8.8	ping an IP address.
ping -n 5 google.com	send 5 ping request
ping -t google.com	ping continously.
ping -l 1000 google.com	send packets of 1000 bytes.
ping -4 google.com	Force IPv4
Ping -6 google.com	force IPV6

Purpose of ping commands:

It checks network connectivity b/w 2 devices & measure response time.

command	purpose.
ping github.com	Ping GitHub.
ping 1.1.1.1	ping cloudflare's public DNS servers.
ping localhost	ping your own computer.
ping -l 1024 google.com	Send 1024 byte packet.
ping -w 1000 google.com	wait up to 1000 milliseconds
Ping 127.0.0.1	Test the loopback interface.
ping -a 8.8.8.8	Resolve the IP address to its hostname, if possible.

esfablishing a ping command :

The ping command works by sending ICMP (Internet Control Message Protocol) Echo Request packet to another device and waiting for an ICMP Echo reply.

Procedure :

* type the CMD command

  ping . google . com .

* DNS lookup

  eg: google.com → 142.250.xx.xx .

* Echo Request is sent.

* Destination receives the request

* Echo reply is sent back.

* Results are displayed.

Ping a website

→ ping google.com

O/P:

pinging google.com [142.250.190.78] with 32 bytes of data:

Reply from 142.250.190.78 : bytes=32 time=18ms TTL=117
Reply from 142.250.190.78 : bytes=32 time=19ms TTL=117
Reply from 142.250.190.78 : bytes=32 time=17ms TTL=117
Reply from 142.250.190.78 : bytes=32 time=18ms TTL=117

Ping statistics for 142.250.190.78 :
Packets : Sent = 4, Received = 4, Lost = 0 (0% loss)

continous Ping

→ ping -t google.com

O/P:

pinging google.com [142.250.190.78] with 32 bytes of data:

Reply from 142.250.190.78 : bytes=32 time=18ms TTL=117
Reply from 142.250.190.78 : bytes=32 time=19ms TTL=117
...
Press ctrl + c to stop.

Result:

Thus ping commands are written & executed successfully

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9. Implimentation of Subnetting Using Cisco Packet Tracker.

AIM:

To implimint classless IP submitting using Cisco Packet Tracer & configure the network devices with suitable IP address & subnet masks.

PROCEDURE:

1. Open CISCO Packet Tracer & create new network topology.
2. Add required routers, switches & PCs.
3. connect devices using appropriate cables.
4. Divide network 192.168.1.0/24 into smaller subnets using /27 subnet mask.
5. configure router interfaces with suitable IP addresses & masks.
6. configure each PC with its IP address, subnet mask, default gateway & DNS information.
7. Test connectivity b/w PCs & routers using ping command.

CODE / COMMANDS:

Router configuration:

```
enable
configure terminal
interface FastEthernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit
```

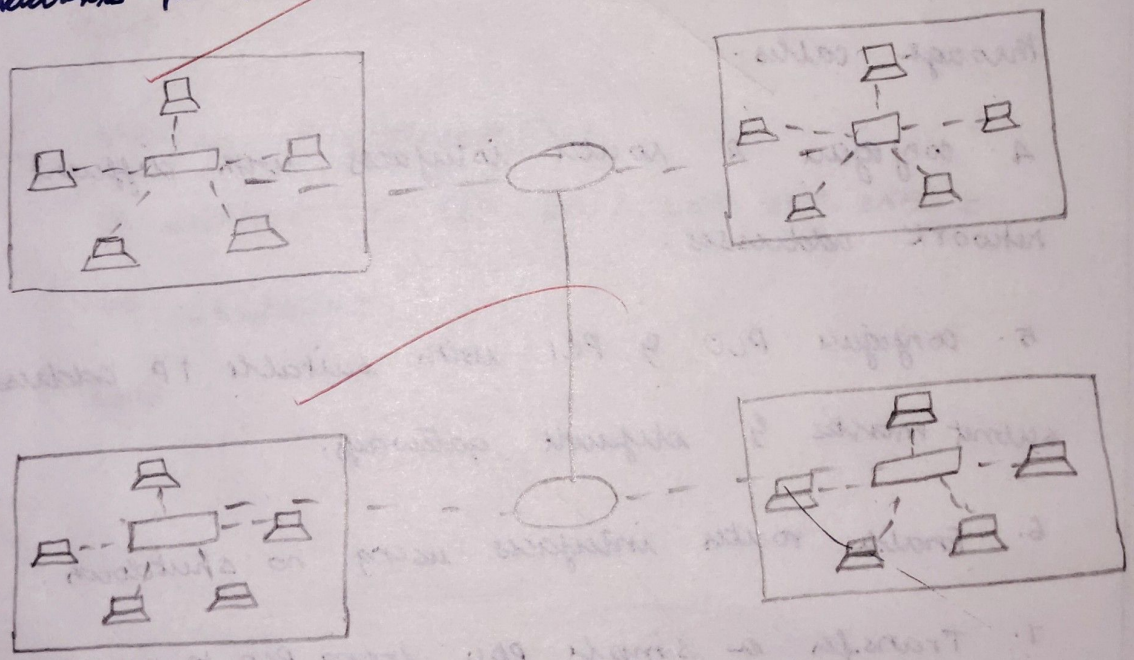
Switch configuration:

```
enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit
interface FastEthernet 0/2
switchport mode access
exit
```

Connectivity Testing:

```
ping 192.168.1.12
ping 192.168.2.12
```

The /27 mask provides 8 subnets with 30 usable host addresses per subnet.



RESULT:

Thus, classless IP subnetting was successfully implemented in CISCO Packet Tracer, & connectivity b/w configured network devices was verified using ping command.

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10. Internetworking using Routers wireless Routers and DHCP.

AIM:

To design & configure a simple internetwork using a routers in CISCO Packet Tracer & establish communication b/w 2 different networks.

PROCEDURE:

1. Open CISCO Packet Tracer
2. Place one routers & 2 PCs in workspace.
3. connect PC0 & PC1 to Router1 using copper straight through cables.
4. Configure 2 router interfaces with different network addresses.
5. Configure PC0 & PC1 with suitable IP addresses, subnet masks & default gateways.
6. Enable routers interfaces using no shutdown.
7. Transfer a simple PDU from PC0 to PC1 to test connectivity.
8. For wireless network section, configure wireless routers with an SSID & enable DHCP.

9. configure laptop to connect wireless network.

10. Configure cisco.com server as a DHCP & DNS server.

11 Use ip config / renew & ping cisco.com to verify connectivity.

CODE / COMMANDS:

Router 1 configuration:

```
enable
config t
interface FastEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface FastEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

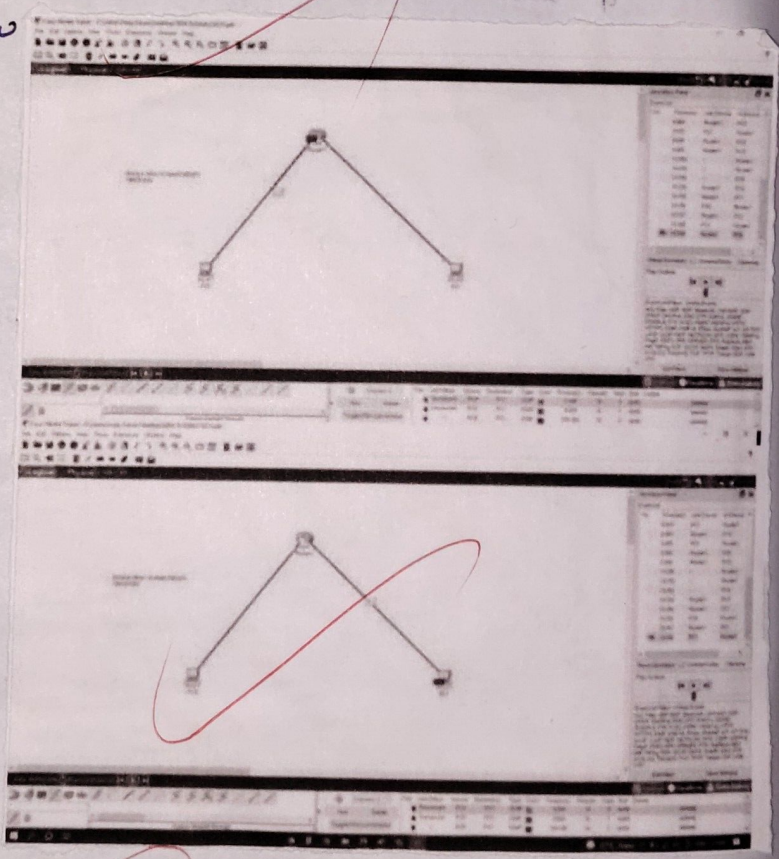
PC configurations:

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

These are addressing values given in lab manual.

Connectivity commands:

ip config /all
ip config /release
ip config /renew
ping cisco.com



RESULT:

Thus, internetwork was successfully designed & configured using routers in Cisco Packet Tracer. communication b/w PC0 & PC1 was verified using PDU listing & DHCP/DNS based connectivity was also tested.

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7. Simple calculator using XML-RPC

AIM:

To develop a simple calculator application using XML-RPC in python.

Requirements:

- Python 3.x 64bit
- Python IDLE
- xmlrpc.server
- xmlrpc.client.

Procedure:

- create Python file named server.py.
- import SimpleXMLRPCServer from xmlrpc.server.
- Define functions for addition, subtraction, multiplication
- Register all calculator func. with server.
- Start server using serve_forever().
- create another file client.py
- import xmlrpc.client & connect server using ServerProxy.
- Get 2 nos & operation choice from user.
- Send selected operation to server.
- The server performs the calculation & returns the result.
- Display the result on client side.

Server Program :

```
from xmlrpc.server import SimpleXMLRPCServer
def add(a,b):
    return a+b
def sub(a,b):
    return a-b
def mul(a,b):
    return a*b
def div(a,b):
    return a/b
server = SimpleXMLRPCServer(("localhost", 8000))
server.register_function(add, "add")
server.register_function(sub, "sub")
server.register_function(mul, "mul")
server.register_function(div, "div")
print("Server is running...")
server.serve_forever()
```

client Program :

```
import xmlrpc.client
server = xmlrpc.client.ServerProxy("http://localhost:8000")
a = float(input(" "))
b = float(input(" "))
print("1. Add")
Print("2. Subtract")
Print("3. Multiply")
Print("4. Divide")
```

```
choice = int(input("Enter choice: "))
if choice == 1:
    print
elif choice == 2:
    print
elif choice == 3:
    print
elif choice == 4:
    print
else:
```

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023,  00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

= RESTART: C:\Users\Dakshinnya\AppData\Local\Programs\Python\Python310\server.py
Server running on port 8000...
>>>
= RESTART: C:\Users\Dakshinnya\AppData\Local\Programs\Python\Python310\client.py
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter first number: 8
Enter second number: 9
Result: 17
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 4
Enter first number: 20
Enter second number: 5
Result: 4.0
--- XML-RPC Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 5
Exiting...
>>>
```

Result :-

Thus, simple calculator using XML-RPC was successfully implemented & executed.

ICMP, TCP / UDP ping scan in TryHackMe platform.

Aim:

To discover live host in TryHackMe target network using Nmap with ARP, ICMP, TCP & UDP host discovery techniques.

Procedure:

Start required Tryhackme machine & connect to the Tryhackme network through the provided VPN.

identify target subnet, for example 10.10.10.0/24.

Perform host discovery using following Nmap scans.

Observe which IP addresses respond & record live hosts.

compare the results obtained using ARP, ICMP, TCP, UDP discovery.

command / code:

1- ARP scan:

```
sudo nmap -sn -PR 10.10.10.0/24.
```

-sn performs host discovery without port scanning & -PR uses ARP requests.

2. ICMP echo scan:

sudo nmap -sn -PE 10.10.10.0/24

-PE sends ICMP echo requests to identify responsive hosts.

3. TCP SYN ping

sudo nmap -sn -PE 10.10.10.0/24

-PE sends ICMP echo requests to identify responsive hosts.

sudo nmap -sN -PS80, 443 10.10.10.0/24

-PS sends TCP SYN probes to specified ports

4. TCP Ack ping

sudo nmap -sN -PA80, 443 10.10.10.0/24

-PA sends TCP Ack probes.

5. UDP ping

sudo nmap -sn -PU53, 161, 10.10.10.0/24

-PU sends UDP probes to specified ports.

Result:

The nmap scans successfully identified the live hosts in target TryHackme network.

13. Demonstrate Network Forensics using PcapXray Tool.

Aim:

To perform network forensics using PcapXray tool by analysing PCAP (Packet capture) file to identify network hosts, communication patterns, protocols etc.

Procedure:

1. Download / obtain sample .pcap file containing captured network traffic.
2. Install required dependencies for PcapXray.
3. Download / clone PcapXray source code from its repo.
4. Open terminal & navigate to PcapXray.
5. Run a tool with sample PCAP file as input.
6. Analyse generated output to identify source & destination IP address.
7. Examine any visualizations / reports to identify unusual patterns.
8. Record observations & summarise findings.

Commands:

* install packages

```
sudo apt update
sudo apt install git python3 python3-pip
```

* Download PcapXray

git clone https://github.com/srivas/PcapXray.git
cd PcapXray.

* Install python dependencies

pip3 install -r requirements.txt.

* Launch Pcap

python3 Source/main.py

* Analyse PCAP file

- select / browse to .pcap file in PcapXray interface
- choose output directory.
- click analyse to process
- click visualise to generate the network diagram.
- open interactive graph to explore communication b/w nodes.

Result:

PcapXray was used to analyse packet capture file & visualise network communications. It can also generate network diagrams & interactive visualizations for further investigation.

14. To capture, save & analyse Network traffic using wireshark.

Aim :

To capture, save & analyse network traffic using wireshark & study protocols TCP, UDP, IP, HTTP, ARP, DHCP ICMP & DNS.

Procedure :

Install & open wireshark.

Select active network interface, such as Wi-fi / Ethernet.

Click start capturing packets to capture network traffic.

Generate network traffic by browsing website, pinging an IP address / performing DNS lookups.

Observe packets captured in real time.

Apply display filters to analyse individual protocols.

Stop capture & save captured packets as a .pcapng file using file → save as.

Analyse packet details to identify source & destination addresses, ports, protocol type & packet information.

Command :

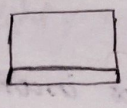
TCP	tcp	connection establishment, data transfer
UDP	udp	connectionless packet transmission
IP	ip	source & destination IPv4 address

HTTP	http	Web requests & responses, when visible.
ARP	arp	IP to mac address resolution.
DHCP	dhcp	IP address allocation & configuration.
ICMP	icmp	ping requests & replies.
DNS	dns	Domain name queries & responses.

useful terminal commands for generating traffic

ping 8.8.8.8

nslookup example.com.



Source Device
generates network packets

Network Interface
wifi / Ethernet

wireshark
captures & analyzes packets

Saved capture file
network traffic. pcapng

Result:

Network traffic was successfully captured, saved, & analysed using wireshark.

12. Develop a program to create shell using TCP Sockets

Aim:

To develop a client-server program using TCP soc to create a simple shell where the client sends command the server & receives.

Procedure:

- Create TCP socket on server using socket()
- Bind server socket to specific IP address & port num using bind().
- Make server listen for incoming connections using lis
- Create TCP socket on client side.
- connect client to server using connect().
- The server receives & executes the command.
- The server sends command output back to client.
- Repeat process until exit is entered.
- close the client & server sockets.

code:

Server Program:

```
import socket
s = socket.socket()
s.bind('localhost', 8080))
s.listen(1)
print("waiting for client")
c, addr = s.accept()
Print("client connected")
msg = c.recv(1024).decode()
Print("client:", msg)
c.send("hello from server".encode())
c.close()
s.close()
```

Client Program:

```
import socket
s = socket.socket()
s.connect(('localhost', 8080))
s.send("Hello Server".encode())
msg = s.recv(1024).decode()
print("Server:", msg)
s.close()
```

Sample Output:

Server Side:

```
waiting for client...
client connected
client: Hello Server
```

Client Side:

```
Server: Hello from Server.
```

Result:

Thus, TCP client-server program was executed successfully & messages were exchanged b/w client & server using TCP sockets.

15. To analyze different types of servers using webalizer TOOL.

Aim:

To analyse web server log files & study server usage statistics such as number of visits, hits, files, pages & bandwidth using webalizer tool.

Procedure:

Install webalizer tool on system.
start the web server such as Apache.
Access different web pages to generate server log entries
Locate webserver access log files.
Run webalizer by providing log file as input.
webalizer analyses log file & generate HTML report
Open generated report in web browser.
Observe statistics such as:

- Hits
- visits
- Pages
- Files
- Bandwidth
- most visited URLs.

Analyse the server activity using generated report.

commands :

sudo apt install webalizer

sudo service apache2 start

webalizer /var/log/apache2/access.log.

To view generated report :

firefox index.html.

Sample Output :

webalizer Report

Total Hits : 150

Total Files : 120

Total Pages : 95

Total Visits : 40

Total Sites : 25

Total Bandwidth : 5.2MB

The webalizer report displays server usage information in form of tables & graphs.

Result :

Thus, the web server log was successfully analyzed using webalizer tool. & various server statistics. such as hits, visits pages, files & bandwidth were studied.